39 — MISTAKE-BOUND MODEL — HALVING ALGORITHM

Aim

To simulate the Halving Algorithm and count the number of prediction mistakes made while learning a target concept from a version space.

Software Required

- Python 3.x
- Jupyter Notebook / Google Colab
- Python Standard Library (math)

Algorithm + Procedure

1. Create a hypothesis space containing threshold hypotheses.
2. Define the target concept.
3. Initialize the version space.
4. Present input examples one at a time.
5. Obtain predictions from the hypotheses.
6. Use majority voting for prediction.
7. Compare the prediction with the true label.
8. If incorrect, count a mistake.
9. Remove inconsistent hypotheses.
10. Repeat for all examples.
11. Calculate $\log_2|H|$.
12. Display the number of mistakes and theoretical bound.

Code

```
import math

# Hypothesis space

H = [

    lambda x, t=t: x >= t

    for t in range(10)

]

# Target concept
```

```python
target = lambda x: x >= 4

version_space = H.copy()

mistakes = 0

# Input examples

examples = list(range(10))

for x in examples:

    votes = [

        h(x)

        for h in version_space

    ]

    majority = (

        sum(votes) > len(votes) / 2

    )

    true_label = target(x)

    if majority != true_label:

        mistakes += 1

        version_space = [

            h for h in version_space

            if h(x) == true_label

        ]

bound = math.log2(len(H))

print("Mistakes made:", mistakes)

print("Mistake bound:", round(bound, 2))
```

Output:

```
Mistakes made: 1
Mistake bound: 3.32
```

Result

The Halving Algorithm was successfully simulated, and the number of mistakes remained within the theoretical bound $\log_2|H|$.